

Problem Statement: Internship Showcase and Tracking Platform

Objective:

Design and develop a **web-based platform** to **showcase internship statistics** across various departments, track the **allotment of faculty mentors**, and monitor **internship progress** for students. This platform will provide **viewers, students, faculty**, and **management** with an intuitive and transparent system for tracking internship activities, mapping them to **Sustainable Development Goals (SDGs)**, **Program Outcomes (PO)**, and **Program Educational Objectives (PEO)** of the course.

The system will allow all stakeholders to visualize internship data, track individual student progress, and assess the alignment of each internship with the institution's academic and sustainability goals.

User Roles & Perspectives

1. Viewers (Public, Employers, Prospective Interns)

Purpose: To explore internship opportunities, view stats across departments, and potentially reach out for collaborations.

Viewer Dashboard Features:

- ✓ **Internship Statistics Overview** – View department-wise internship stats, including number of internships, type of companies, and sectors.
 - ✓ **Internship Details** – View available internships with descriptions, requirements, and SDG/PO/PEO alignment.
 - ✓ **Search & Filters** – Filter internships by **department**, **SDG**, **PO/PEO** mapping, or **industry sector**.
 - ✓ **Project Outcomes** – View how internships contribute to specific **Program Outcomes (PO)** and **Program Educational Objectives (PEO)**.
 - ✓ **Contact Internship Providers** – Reach out to companies or faculty coordinators for collaboration or networking opportunities.
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2. Students (Internship Seekers)

Purpose: To track internship opportunities, apply for internships, and monitor the status of their internship with faculty mentorship.

Student Dashboard Features:

- ✓ **Available Internship List** – View and filter available internships based on department, company, SDGs, and PO/PEO.
 - ✓ **Application Tracking** – Track the status of applications, including approval or rejection, and deadlines.
 - ✓ **Mentor Allotment** – View assigned **faculty mentor** for guidance during the internship, along with mentor contact details.
 - ✓ **Internship Progress Tracker** – Track the progress of internship activities, deadlines, tasks completed, and learning outcomes.
 - ✓ **SDG, PO, and PEO Mapping** – See how the internship relates to the **Sustainable Development Goals (SDGs)** and how it fulfills **Program Outcomes (PO)** and **Program Educational Objectives (PEO)**.
 - ✓ **Feedback & Evaluation** – Receive feedback from faculty mentors and industry supervisors on internship performance.
 - ✓ **Reports & Analytics** – Generate reports that track personal growth, skill development, and SDG contributions throughout the internship.
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3. Faculty Coordinators (Mentors, Course Instructors)

Purpose: To assign mentors to students, track the progress of internships, and evaluate student performance based on **SDGs**, **POs**, and **PEOs**.

Faculty Coordinator Dashboard Features:

- ✓ **Internship Overview** – View the list of all students, the internships they are pursuing, and the companies involved.
 - ✓ **Mentorship Assignment** – Assign faculty mentors to individual students or groups based on their interests, department, or specialization.
 - ✓ **Internship Progress Monitoring** – Monitor student progress through internship status updates, milestones, and task completion.
 - ✓ **SDG/PO/PEO Alignment Review** – Ensure that the internship activities are aligned with the **SDGs**, **Program Outcomes (POs)**, and **Program Educational Objectives (PEOs)**.
 - ✓ **Student Feedback & Evaluation** – Provide mentorship feedback and evaluate internship performance based on industry experience and academic criteria.
 - ✓ **Collaboration with Industry Supervisors** – Engage with industry mentors to assess student performance and suggest improvements.
 - ✓ **Reports & Analytics** – Generate reports that track internship success rates, student progress, and alignment with academic objectives.
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4. Admins (System Moderators, Platform Administrators)

Purpose: To manage internship listings, assign roles, track internship statistics, and generate reports.

Admin Dashboard Features:

- ✓ **Internship Listing Management** – Manage internship opportunities by approving, editing, or removing internship postings.
 - ✓ **Role Management** – Assign roles to students, faculty, and industry mentors, ensuring appropriate access for each.
 - ✓ **Internship Stats** – View department-wise internship data, including student participation rates, industry partners, and SDG alignment.
 - ✓ **Progress Monitoring** – Monitor the overall progress of internships across all departments, including individual student progress.
 - ✓ **Notifications & Alerts** – Send notifications for upcoming deadlines, internship applications, and faculty/mentor assignments.
 - ✓ **SDG, PO, and PEO Mapping Oversight** – Ensure all internships are properly mapped to SDGs, POs, and PEOs.
 - ✓ **Reports & Insights** – Generate comprehensive reports on internship statistics, student performance, SDG contributions, and department-specific trends.
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5. Management (Institutional Coordinators, University Officials)

Purpose: To oversee internship programs at an institutional level, assess the effectiveness of faculty mentorship, and ensure the alignment of internships with academic and sustainability goals.

Management Dashboard Features:

- ✓ **Institution-Wide Internship Overview** – View the total number of internships, student participation rates, and industry collaboration statistics.
 - ✓ **SDG Contribution Tracking** – Track how internships contribute to institutional goals related to **Sustainable Development Goals (SDGs)**, **Program Outcomes (POs)**, and **Program Educational Objectives (PEOs)**.
 - ✓ **Program Success Metrics** – Evaluate the success of the internship program based on student feedback, performance, and industry collaboration.
 - ✓ **Internship Placement Analytics** – Review how successful students are in securing internships, and monitor trends in placements by industry.
 - ✓ **Report Generation** – Generate detailed reports on internship performance, SDG alignment, and department-level analytics.
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Key Functional Requirements for the Platform

✓ Internship Listing & Management

- Admins and faculty can post, edit, and approve internship opportunities from industry partners.
- Internships can be filtered by department, industry sector, SDG, PO/PEO mapping, and availability.
- Students can apply for internships, track application status, and receive notifications for updates.

✓ SDG, PO, and PEO Mapping

- **Students:** Map their internships to relevant SDGs and academic outcomes (POs and PEOs).
- **Faculty:** Ensure proper alignment of student internships with the course's SDGs, POs, and PEOs.
- **Admins:** Review and approve the SDG and PO/PEO mapping for each internship.

✓ Internship Progress Tracking

- Students can track their internship progress with milestones, deadlines, and task completion.
- Faculty can monitor the overall progress of students and provide feedback.

✓ Mentorship Assignment & Communication

- **Faculty** assign mentors to students based on their skills and internship requirements.
- Students and mentors can communicate directly within the platform for updates and feedback.

✓ Feedback & Evaluation

- Faculty mentors and industry supervisors provide regular feedback to students on performance and achievements.
- Students can review and act on feedback to improve their internship outcomes.

✓ Internship Reports & Analytics

- **Admins** and **faculty** can generate reports on student performance, internship success, SDG contributions, and overall program effectiveness.
- **Management** can evaluate the internship program's alignment with academic and sustainability goals.

✓ User Authentication & Role-Based Access

- Secure login system with role-based access for students, faculty, and admins.
 - Faculty mentors can view and manage student internship progress, while admins have full control over internship listings and user roles.
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Technology Stack (Suggested for Full Stack Development Course)

- **Frontend:** React.js / Next.js (for building dynamic and interactive UIs)
 - **Backend:** Node.js (Express.js) / Django (for API development)
 - **Database:** MongoDB (for flexible data storage) or MySQL (for structured data)
 - **Authentication:** Firebase Auth / OAuth (for secure login)
 - **Cloud Storage:** AWS S3 (for storing internship-related media files and documents)
 - **Hosting:** AWS / Vercel / Netlify (for deployment)
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